

Project Proposal

Project Title: SMS Spam Classifier using Machine Learning

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1. Project Overview

- **Project Topic:**

This project focuses on developing an AI-based SMS Spam Classifier that can automatically detect whether a text message is spam or legitimate (ham). With the increasing use of mobile communication, users frequently receive unwanted promotional or fraudulent messages. The system will use machine learning and natural language processing techniques to analyze SMS text and classify it accordingly.

- **Objective:**

The main objective of this project is to design and implement an intelligent system capable of automatically identifying spam messages. The system aims to improve user experience by filtering unwanted messages, reducing fraud risks, and demonstrating the practical use of AI techniques such as text preprocessing, feature extraction, and classification algorithms.

2. System Description

- **Original Problem Background:**

Spam messages are a common problem in mobile communication. These messages often contain advertisements, phishing attempts, or fraudulent offers. Manually identifying spam messages is inefficient and time-consuming for users. Therefore, an automated system is needed that can quickly analyze message content and determine whether it is spam or not.

- **Innovations Introduced:**

The proposed system introduces an AI-based approach to automatically classify SMS messages. Natural Language Processing (NLP) techniques will be used to clean and preprocess the text, and machine learning algorithms will analyze patterns in spam

messages. The system can also be integrated with messaging platforms or applications to automatically filter spam in real-time.

3. AI Approach and Methodology

- **Search Algorithms:**

Search algorithms are not the main focus of this project; however, optimization techniques may be used during model training to improve classification accuracy.

- **LLM / OpenAI API / Other AI APIs:**

Natural language processing techniques will be used to analyze SMS content. Pre-trained language models or APIs may assist in understanding text patterns and improving classification.

- **Automation Tools:**

Automation tools may be used to process incoming messages and automatically classify them as spam or ham.

- **Other Techniques:**

The project will use text preprocessing, tokenization, stop-word removal, and feature extraction techniques such as Bag-of-Words or TF-IDF. Machine learning algorithms such as Naive Bayes or Logistic Regression will be used for classification.

4. System Rules and Mechanics

- **Modified Rules / System Workflow:**

User provides SMS message → Text preprocessing → Feature extraction → Machine learning model processes the input → Message classified as Spam or Ham → Result displayed to the user.

- **Output / Success Conditions:**

The system successfully classifies incoming SMS messages with high accuracy. The output will clearly indicate whether the message is spam or legitimate.

- **Process Sequence:**

1. Input SMS message
2. Clean and preprocess text
3. Convert text into numerical features
4. Apply trained machine learning model

5. Predict spam or ham
6. Display classification result

5. Libraries and Tools

- OpenAI API / Other LLM APIs
- Python Libraries: NumPy, Pandas, Scikit-learn, NLTK
- Machine Learning Libraries: Scikit-learn
- Other Tools: Jupyter Notebook / Python IDE

6. References

1. [SMS-Text Classification](#) – Github repository
2. [SPAM classifier-how to build](#) – YouTube video